The Mole Concept

Moles, molar mass and Avogadro's number

Name _____ Class _____ Date _____

DATA

H = 1.008 g/mol C = 12.01 g/mol N = 14.01 g/mol O = 16.00 g/mol Na = 22.99 g/mol

Mg = 24.31 g/mol Cl = 35.45 g/mol Ca = 40.08 g/mol

Avogadro constant: 6.022×10^{23} mol^{-1}. Show your working and give answers to an appropriate number of significant figures.

A. Understanding the mole

1. What is a mole? Answer in one sentence.

2. One mole of carbon has a mass of 12.01 g, but one mole of magnesium has a mass of 24.31 g. Do the two samples contain the same number of atoms? Explain.

3. A student says “one mole of oxygen gas contains 6.022×10^{23} oxygen atoms.” What is wrong with this statement?

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B. Molar mass

4. Calculate the molar mass of each substance.

(a) methane, CH_4 (b) ammonia, NH_3 (c) calcium carbonate, $CaCO_3$

5. Calculate the molar mass of (a) glucose, $C_6H_{12}O_6$, and (b) magnesium chloride, $MgCl_2$.

C. Converting between mass, moles and particles

6. How many moles are in 25.0 g of methane, CH_4?

7. What is the mass of 0.150 mol of calcium carbonate, $CaCO_3$?

8. How many molecules are in 0.500 mol of ammonia, NH_3?

9. How many moles of water are in a sample containing 1.50×10^{24} water molecules?

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10. What is the mass of 3.01×10^{23} atoms of sodium?

D. Counting atoms inside compounds

11. How many moles of hydrogen atoms are in 2.00 mol of glucose, $C_6H_{12}O_6$?

12. How many oxygen atoms are in 10.0 g of carbon dioxide, CO_2?

13. What mass of chlorine is contained in 1.00 mol of magnesium chloride, $MgCl_2$?

E. Challenge

14. Which contains more molecules: 10.0 g of water, H_2O, or 10.0 g of carbon dioxide, CO_2? Show your reasoning.

15. A 5.00 g sample of a metallic element contains 0.125 mol of atoms. Calculate the molar mass of the element and use the data box to identify it.

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